Sales Data Analysis Report

1. Total Revenue

This analysis begins by calculating the **total revenue generated by the company** during the analyzed period.

The results show that the company produced **strong overall revenue**, indicating consistent sales activity.

```
(Super+Enter) = pd.DataFrame(data)

total_revenue = df["Revenue"].sum()
print("Total Revenue:", round(total_revenue, 2))

17]  ✓ 0.0s                                  Pytho

Total Revenue: 4620540.41
```

2. Average Revenue per Transaction

The average revenue per transaction measures how much money the company earns **from each individual sale**.

This KPI helps evaluate:

- Customer spending behavior
- Pricing strategy effectiveness
- Overall transaction value

```python
df = pd.DataFrame(data)

average_order_value = df["Revenue"].mean()
print("Average Order Value:", round(average_order_value, 2))
```

✓ 0.0s

```
Average Order Value: 924.11
```

3. Top Revenue-Generating Products

A product-level analysis was conducted to determine **which products generate the highest revenue**.

Understanding top-performing products helps the company:

- Focus marketing efforts
- Improve inventory management
- Maximize profitability

```python
df = pd.DataFrame(data)

revenue_by_product = (
    df.groupby("Product")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

print(revenue_by_product)
```

9] ✓ 0.0s

```
Product
Laptop        1541101.84
Smartphone     923434.35
Tablet         679252.29
Monitor        451637.85
Desk Chair     354478.37
Printer        303346.42
Headphones     179868.65
Keyboard       100664.09
Mouse           60683.60
USB Cable       26072.95
Name: Revenue, dtype: float64
```

4. Revenue by Category

Revenue was also analyzed across **product categories** to identify which groups of products contribute the most to total sales.

This analysis helps the business understand:

- Which product groups perform best
- Where demand is strongest
- Which categories may require improvement

```python
df = pd.DataFrame(data)

revenue_by_category = (
    df.groupby("Category")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

print(revenue_by_category)
```

✓ 0.0s

```
Category
Electronics    3595426.33
Accessories     367289.29
Furniture       354478.37
Office          303346.42
Name: Revenue, dtype: float64
```

5. Revenue by Region

Regional analysis examines how revenue is distributed **across different geographic areas**.

This helps identify:

- High-performing markets
- Areas with growth potential
- Regional differences in customer demand

```python
df = pd.DataFrame(data)

revenue_by_region = (
    df.groupby("Region")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

print(revenue_by_region)
```

✓ 0.0s

```
egion
antiago          1018152.13
an Pedro          916523.32
a Romana          912592.61
anto Domingo      903827.97
uerto Plata       869444.38
ame: Revenue, dtype: float64
```

6. Regional Revenue Visualization

A regional revenue chart was created to visually compare **sales performance across regions**.

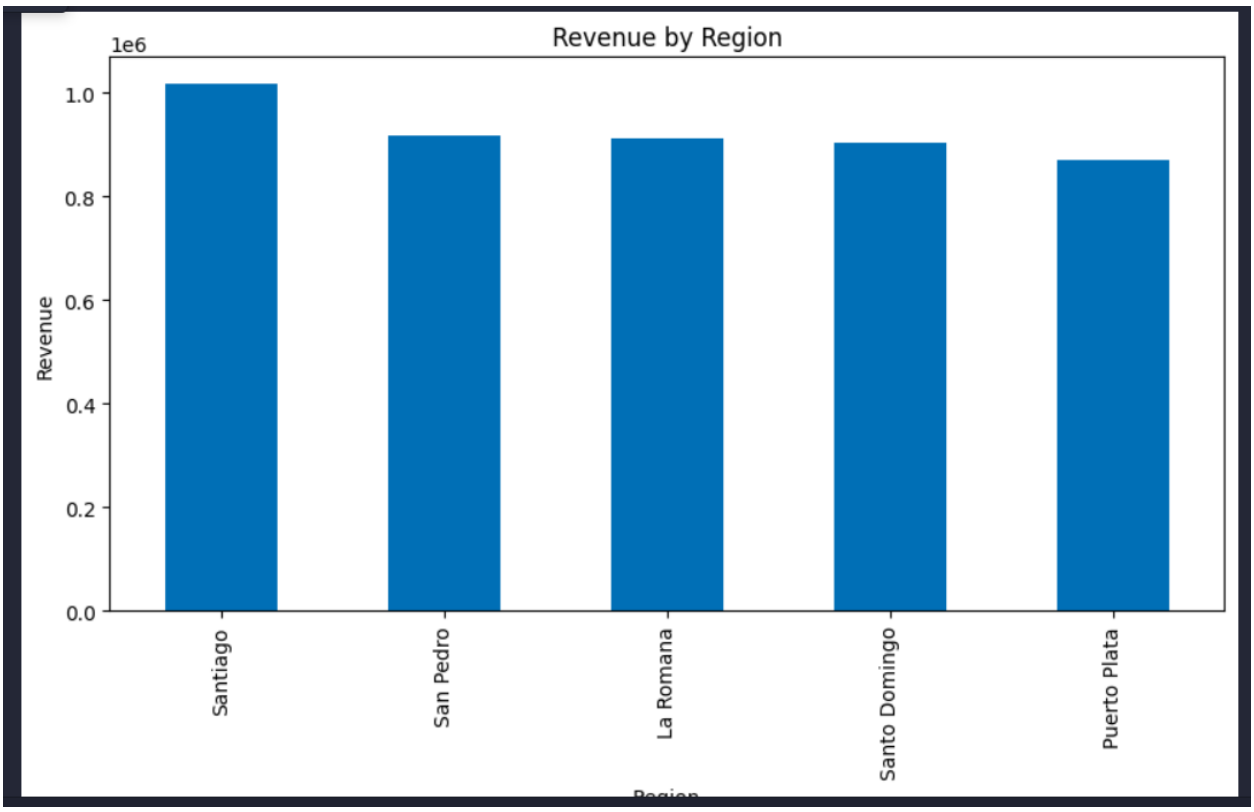
This chart allows stakeholders to quickly identify:

- The strongest markets
- Regions with lower sales performance

```python
revenue_by_region.plot(
    kind="bar",
    figsize=(10,5),
    title="Revenue by Region"
)

plt.xlabel("Region")
plt.ylabel("Revenue")

plt.show()
```



7. Revenue Calculation

Revenue was calculated using the following formula:

Revenue = Quantity × Price

This calculation provides the foundation for all further analysis within the report.

```python
df["Month"] = df["Order_Date"].dt.to_period("M")

monthly_revenue = df.groupby("Month")["Revenue"].sum()

print(monthly_revenue)
```

```
Month
2023-01    187411.91
2023-02    176969.10
2023-03    217443.27
2023-04    202247.13
2023-05    195733.16
2023-06    198530.43
2023-07    172915.40
2023-08    207017.27
2023-09    179741.83
2023-10    211216.60
2023-11    201221.38
2023-12    172932.59
2024-01    197061.27
2024-02    187095.03
2024-03    207063.14
2024-04    160885.67
2024-05    186952.57
2024-06    170448.83
2024-07    213686.63
2024-08    199302.75
2024-09    199180.35
2024-10    204844.18
2024-11    178624.92
2024-12    192015.00
Freq: M, Name: Revenue, dtype: float64
```

8. Revenue Trend Analysis

A revenue trend chart was created to visualize **sales performance over time**.

This allows us to observe:

- Growth patterns

- Seasonal fluctuations
- Potential business cycles

```python
monthly_revenue.plot(
    kind="line",
    figsize=(10,5),
    title="Monthly Revenue Trend"
)

plt.xlabel("Month")
plt.ylabel("Revenue")

plt.show()
```

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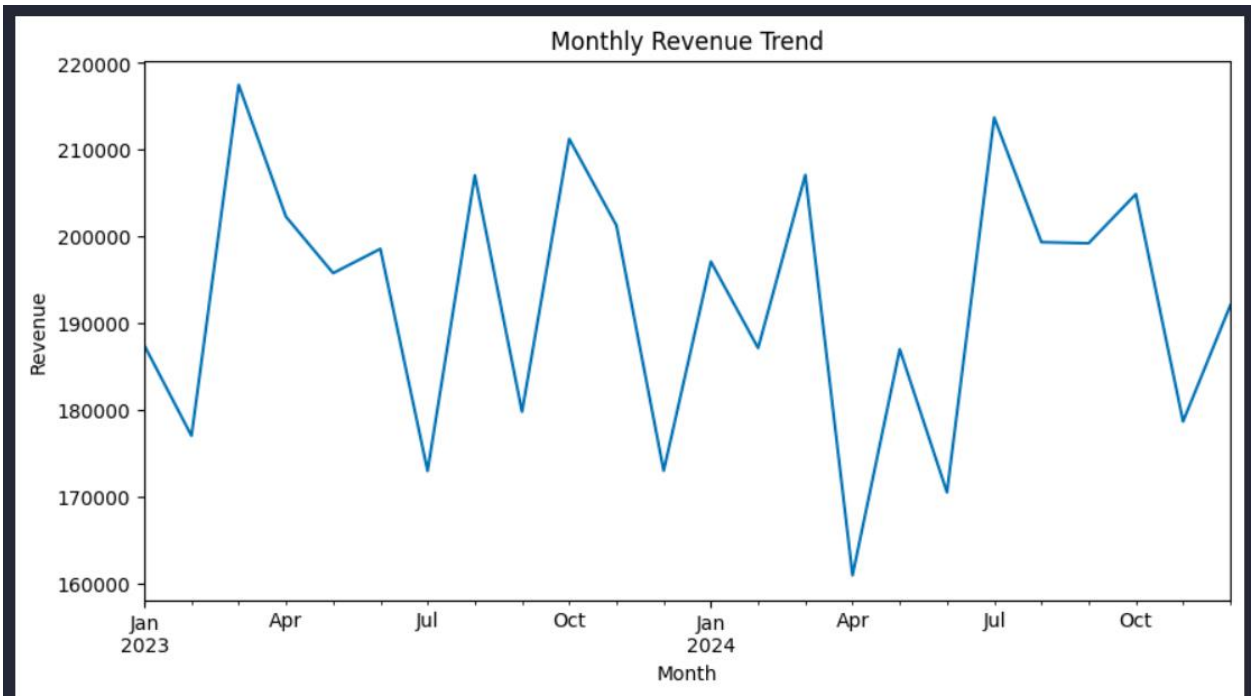


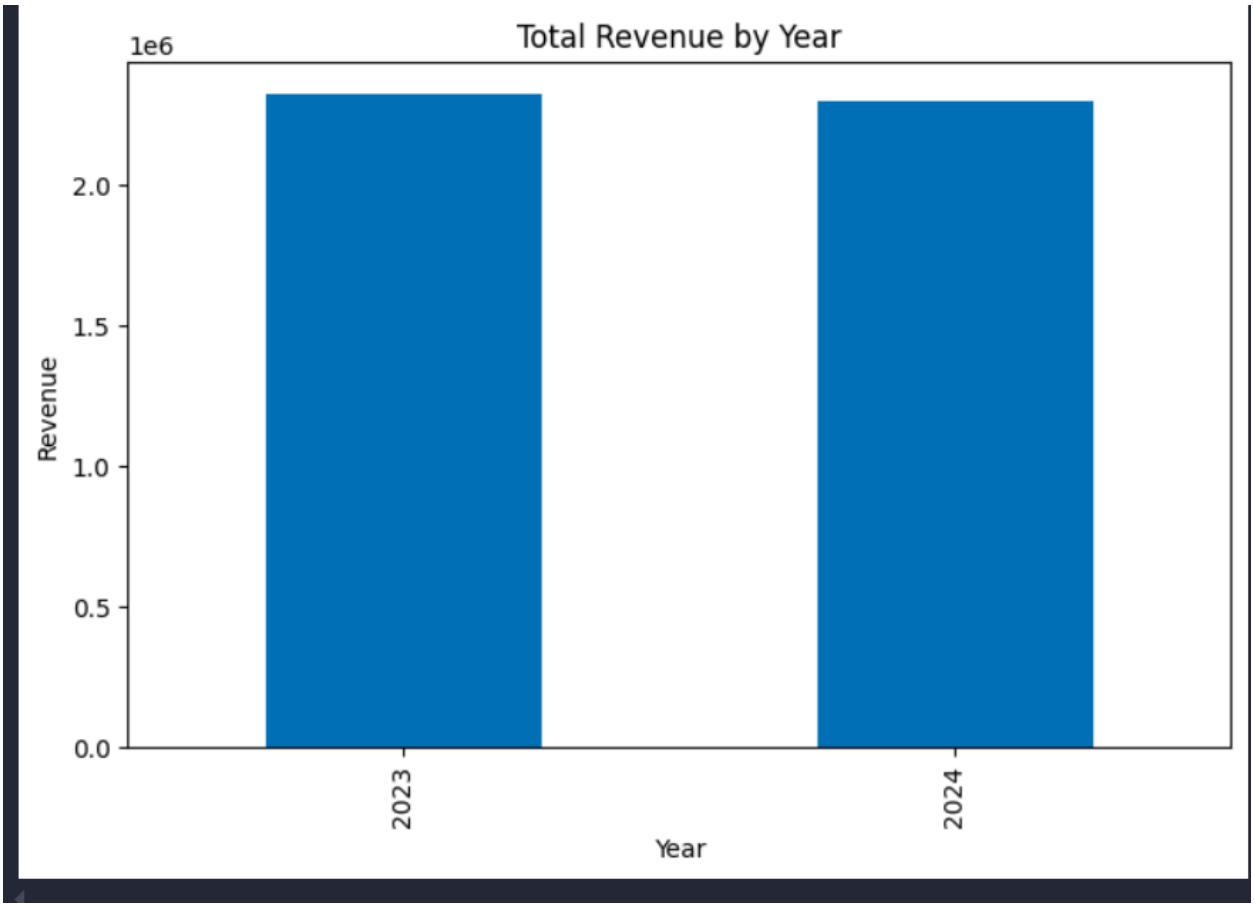
9. Revenue by Year

Insight

Total revenue remained relatively stable between **2023 and 2024**, with both years generating approximately **$2.3 million in sales**.

This suggests:

- Consistent customer demand
- A stable market position
- Limited year-over-year growth



Insight

Total revenue remained relatively stable between 2023 and 2024, with both years generating approximately $2.3M in sales. This suggests consistent demand but limited year-over-year growth.

10. Revenue by Product

A product-level revenue analysis identifies how much each product contributes to overall company sales.

This helps reveal:

- Top-selling products
- Low-performing items
- Opportunities for product strategy improvements

```
revenue_by_product = (
    df.groupby("Product")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

print(revenue_by_product)
```

✓ 0.0s

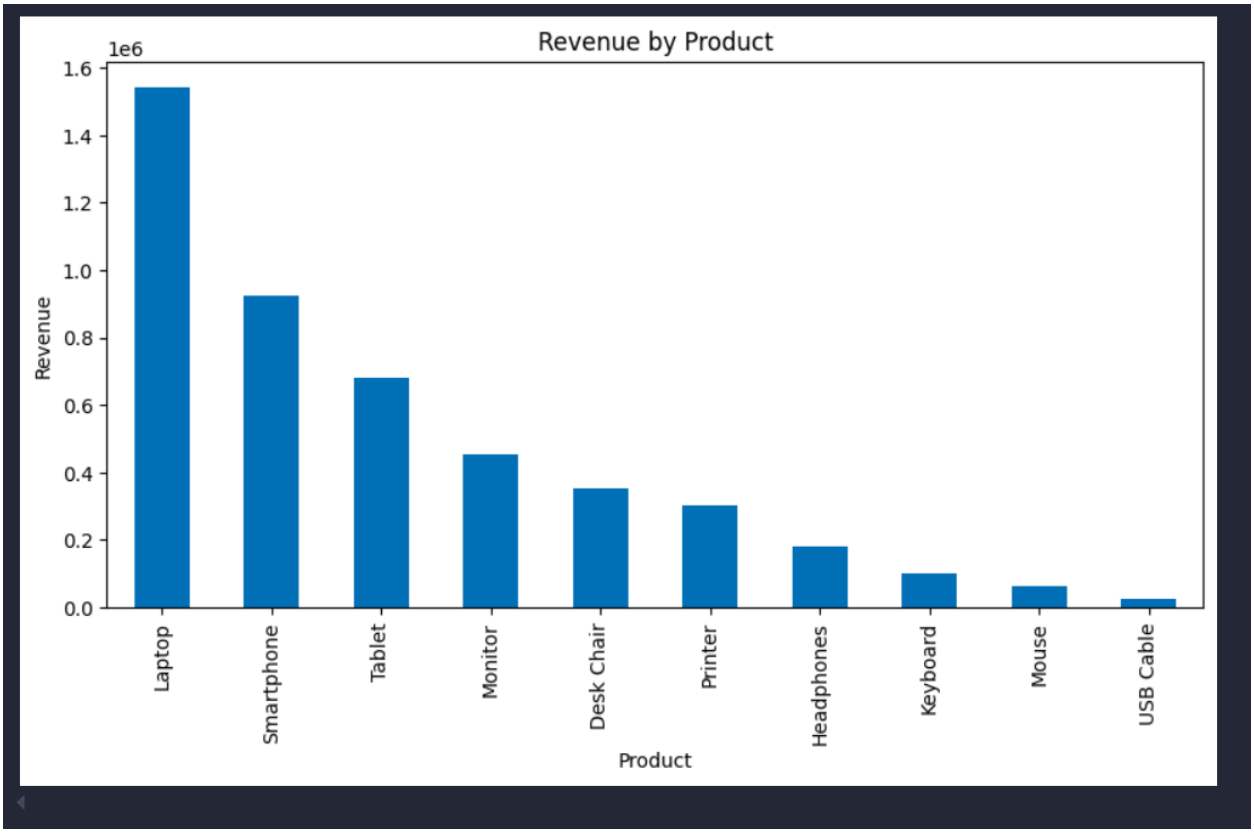
```
Product
Laptop        1541101.84
Smartphone     923434.35
Tablet         679252.29
Monitor        451637.85
Desk Chair     354478.37
Printer        303346.42
Headphones     179868.65
Keyboard       100664.09
Mouse           60683.60
USB Cable       26072.95
Name: Revenue, dtype: float64
```

11. Top Product Chart

A visualization was created to highlight the **top-performing products** based on revenue contribution.

This allows decision-makers to quickly identify which products drive the business.

```
revenue_by_product.plot(
    kind="bar",
    figsize=(10,5),
    title="Revenue by Product"
)

plt.xlabel("Product")
plt.ylabel("Revenue")

plt.show()
```



12. Pareto Analysis

A Pareto analysis was conducted to determine whether **a small number of products generate the majority of revenue**.

This analysis follows the **80/20 principle**, which suggests that:

- Roughly 80% of results come from 20% of the causes.

```
pareto = revenue_by_product.reset_index()

pareto["Cumulative_Revenue"] = pareto["Revenue"].cumsum()
pareto["Cumulative_Percentage"] = pareto["Cumulative_Revenue"] / pareto["Revenue"].sum() * 100

print(pareto)
```

✓ 0.0s

```
    Product     Revenue  Cumulative_Revenue  Cumulative_Percentage
     Laptop  1541101.84          1541101.84              33.353281
 Smartphone   923434.35          2464536.19              53.338700
     Tablet   679252.29          3143788.48              68.039411
    Monitor   451637.85          3595426.33              77.813979
 Desk Chair   354478.37          3949904.70              85.485773
    Printer   303346.42          4253251.12              92.050945
 Headphones   179868.65          4433119.77              95.943751
   Keyboard   100664.09          4533783.86              98.122372
      Mouse    60683.60          4594467.46              99.435716
  USB Cable    26072.95          4620540.41             100.000000
```

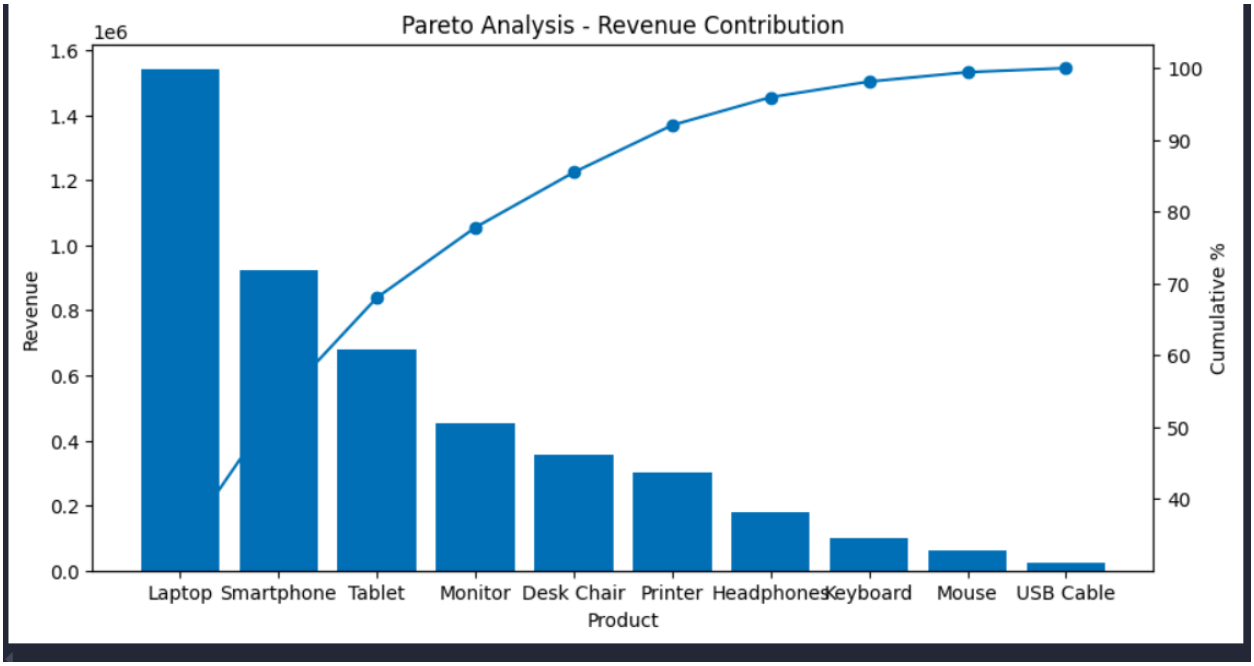
13. Pareto Chart

The Pareto chart visually shows:

- Individual product contributions
- Cumulative revenue percentage

This helps identify **which products have the greatest impact on revenue**.

```python
fig, ax = plt.subplots(figsize=(10,5))

ax.bar(pareto["Product"], pareto["Revenue"])
ax2 = ax.twinx()

ax2.plot(pareto["Product"], pareto["Cumulative_Percentage"], marker="o")

ax.set_xlabel("Product")
ax.set_ylabel("Revenue")
ax2.set_ylabel("Cumulative %")

plt.title("Pareto Analysis - Revenue Contribution")

plt.show()
```
✓ 0.5s



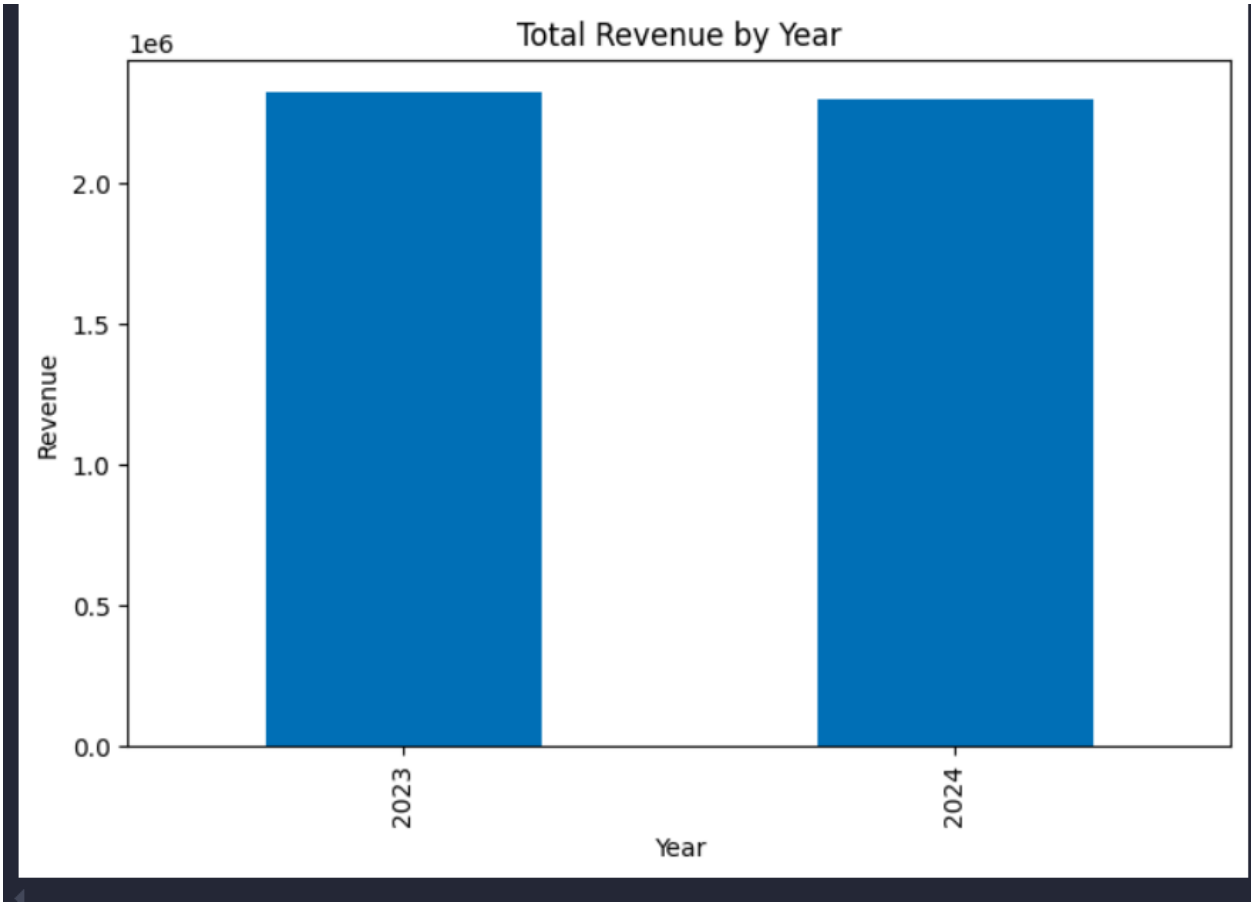
Business Insights and Recommendations

Overall Revenue Performance

The company generated strong revenue across the analyzed period. Total revenue remained **stable between 2023 and 2024**, with both years producing similar results.

While growth was limited, the stability suggests:

- A **consistent customer base**
- Reliable product demand



Seasonal and Monthly Sales Trends

The monthly trend analysis revealed **noticeable fluctuations throughout the year**, indicating seasonal purchasing behavior.

Certain months consistently generate higher revenue, which may correspond to:

- Promotional periods
- Holiday seasons
- Peak demand cycles

Understanding these patterns can help the company **better plan marketing campaigns and inventory levels**.

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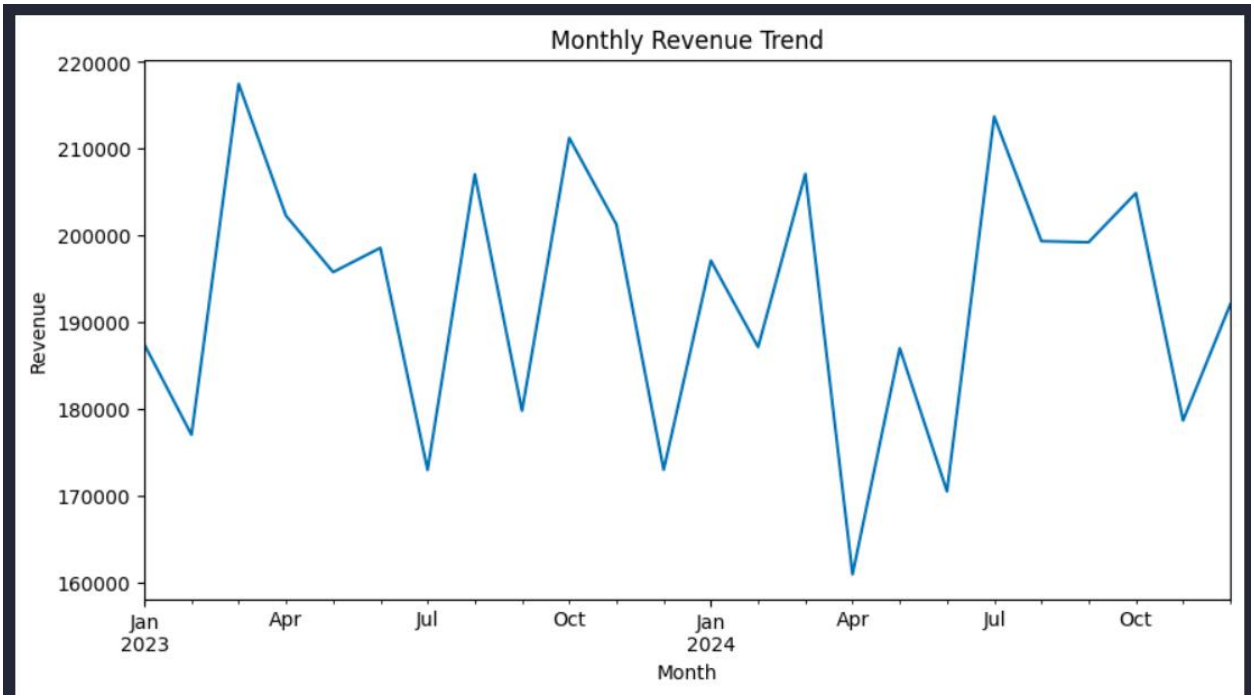


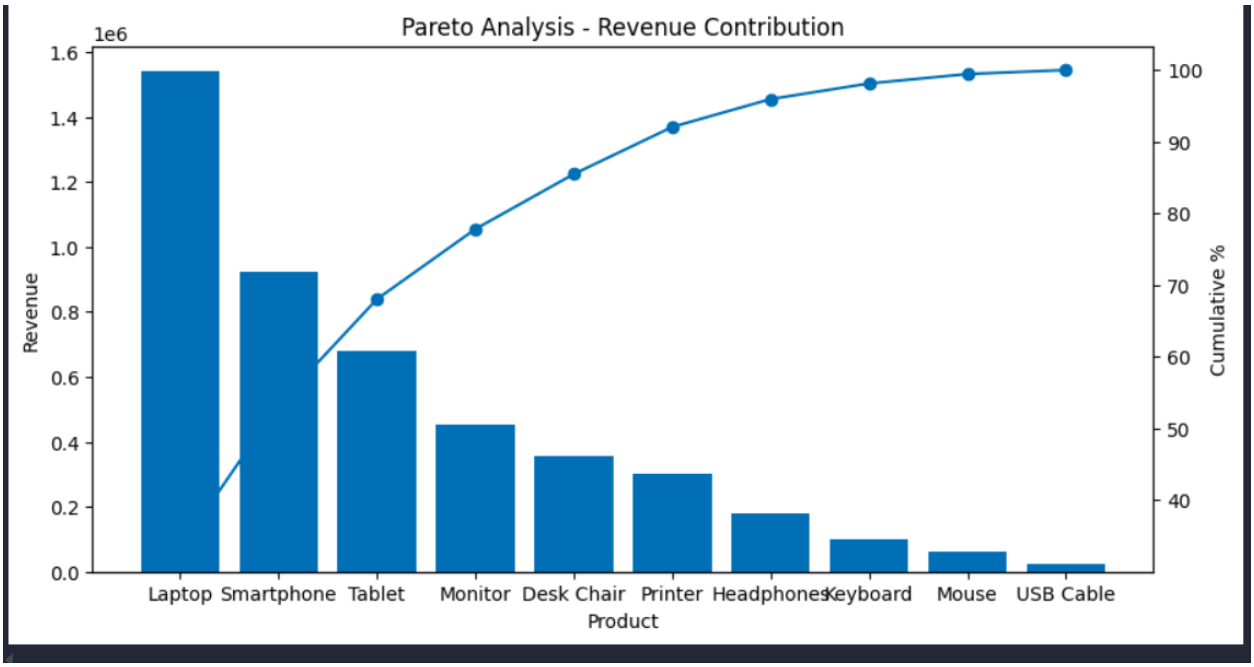
Product Contribution Analysis

The Pareto analysis shows that **a small number of products generate a large percentage of total revenue**.

This indicates that company revenue is **highly concentrated among top-performing products**.

Focusing on these key products can help increase profitability through:

- Targeted marketing
- Strategic inventory management
- Improved product availability



Regional Sales Performance

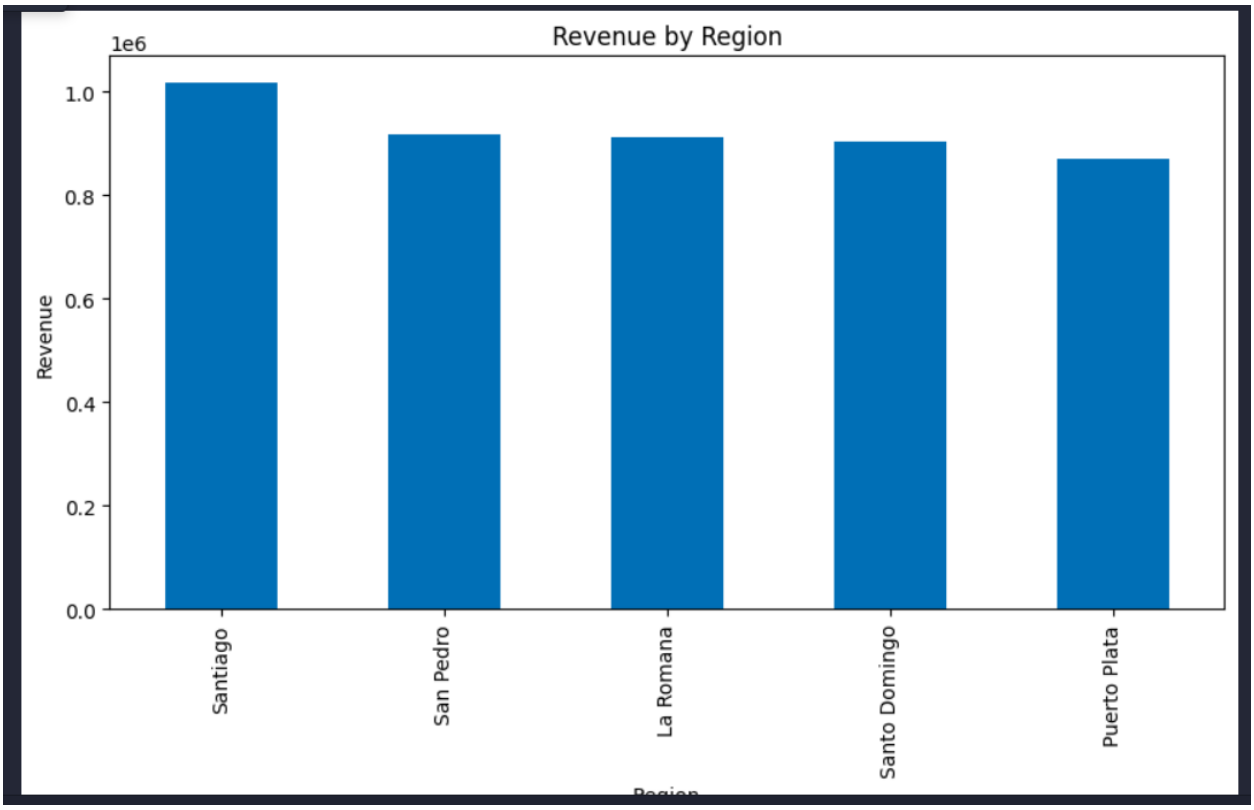
Regional analysis reveals that **certain regions contribute significantly more revenue than others**.

Major metropolitan areas generate the highest sales, suggesting:

- Stronger market penetration
- Higher customer demand

Meanwhile, regions with lower sales may represent **untapped growth opportunities**.

Expanding marketing efforts or improving distribution strategies in these areas could increase overall revenue.



Strategic Recommendations

Based on the analysis, several opportunities for business improvement were identified:

1. **Prioritize high-performing products** that generate the majority of revenue.
2. **Align marketing campaigns with seasonal demand peaks** to maximize sales during high-revenue months.
3. **Expand marketing and sales strategies in underperforming regions** to capture new market opportunities.
4. **Monitor year-over-year trends** to identify growth opportunities and potential revenue risks.